

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
DLA 17

Regulation of Breathing

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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0227	Identify the regions in the central nervous system that play important roles in the generation and control of cyclic breathing.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0228	Give three examples of reflexes involving pulmonary receptors that influence breathing frequency and tidal volume. Describe the receptors and neural pathways involved.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0229	List the anatomical locations of chemoreceptors sensitive to changes in arterial PO ₂ , PCO ₂ , and pH that participate in the control of ventilation.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0372	Differentiate the relative importance of each in sensing alterations in blood gases.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0230	Describe how changes in arterial PO ₂ and PCO ₂ alter alveolar ventilation, including the synergistic effects when PO ₂ and PCO ₂ both change.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0231	Explain the respiratory drive in a COPD patient.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0373	Explain the change in respiratory drive when oxygen is given to a COPD patient.

Recommended Reading

1. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell. Chapter 21: Control of Breathing.
2. Costanzo Physiology , 7e. Linda S.Costanzo. Chapter 5: Section : Control of Breathing.

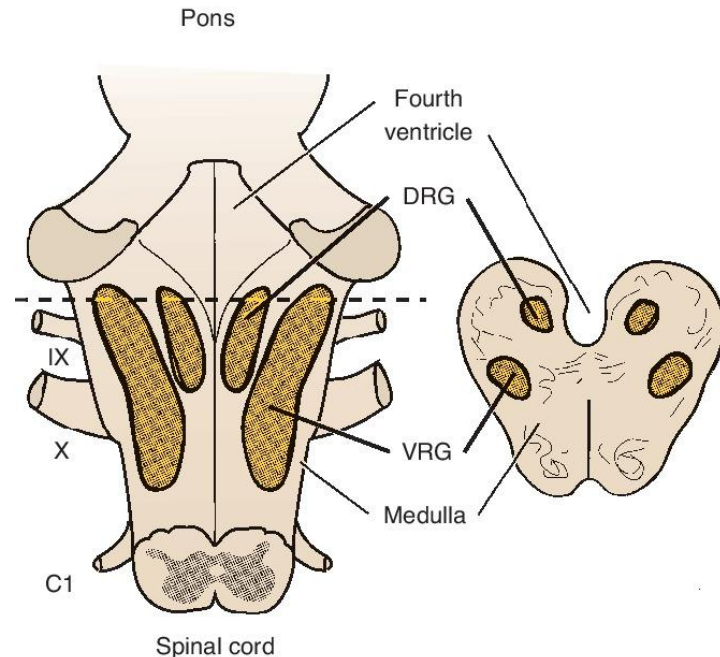
Please Note:

- This DLA is NOT accompanied by a video.
- If you need more explanation of the information contained in this DLA, you are encouraged to read material from the above recommended readings.

Generation & Control of Breathing – Role of CNS

Neural Control of Breathing

- Undisputed Fact : Medulla oblongata in the brain has the respiratory rhythm generator
- The dorsal respiratory group (DRG) and the ventral respiratory group (VRG) are located in the medulla
- DRG and parts of rostral VRG have neurons that drive inspiration
- Pontine Nuclei : Act as a switch between Inspiration & Expiration
- Most important intrinsic respiratory rhythm generator in the Medulla is called preBötzinger complex
- Non-rhythmogenic pontine areas like pneumotaxic center and apneustic center inputs to medulla to control smooth respiratory rhythm



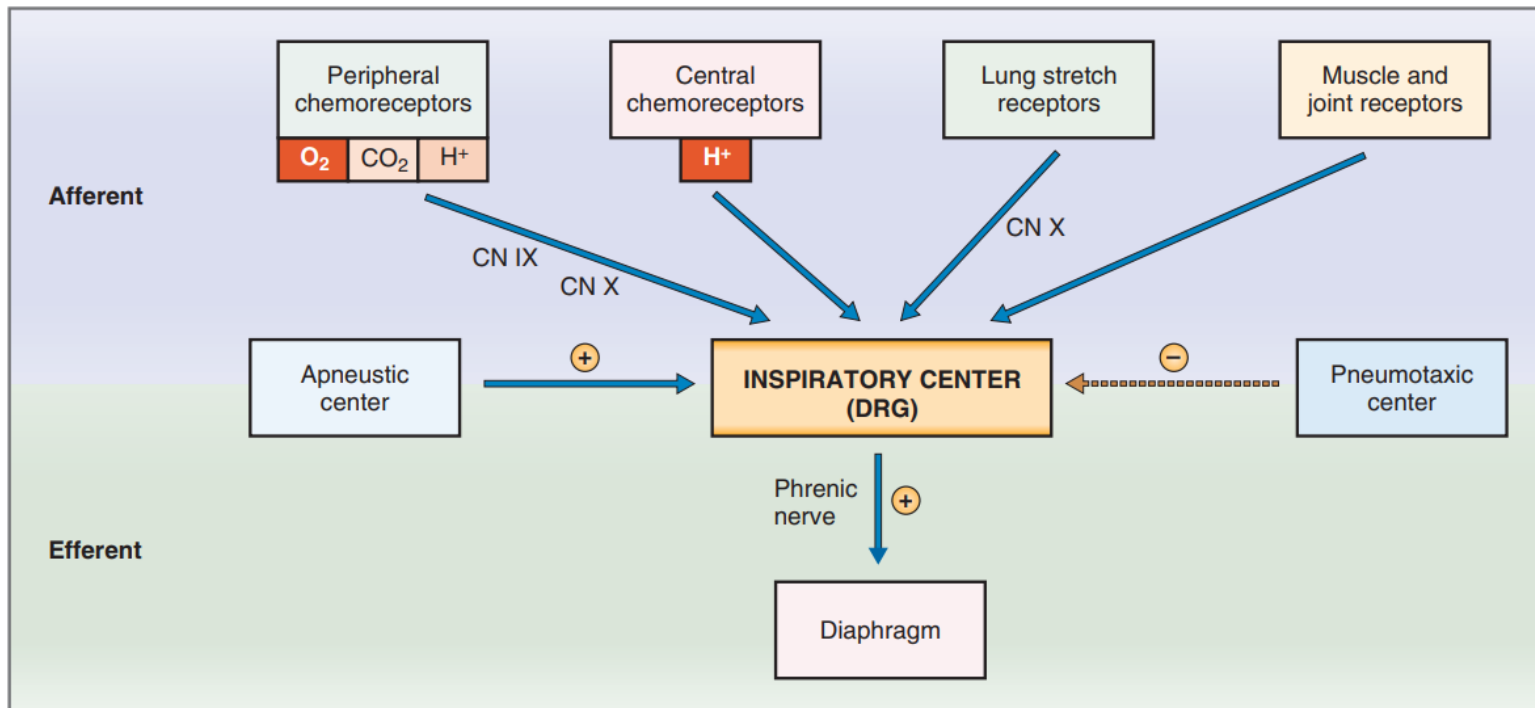
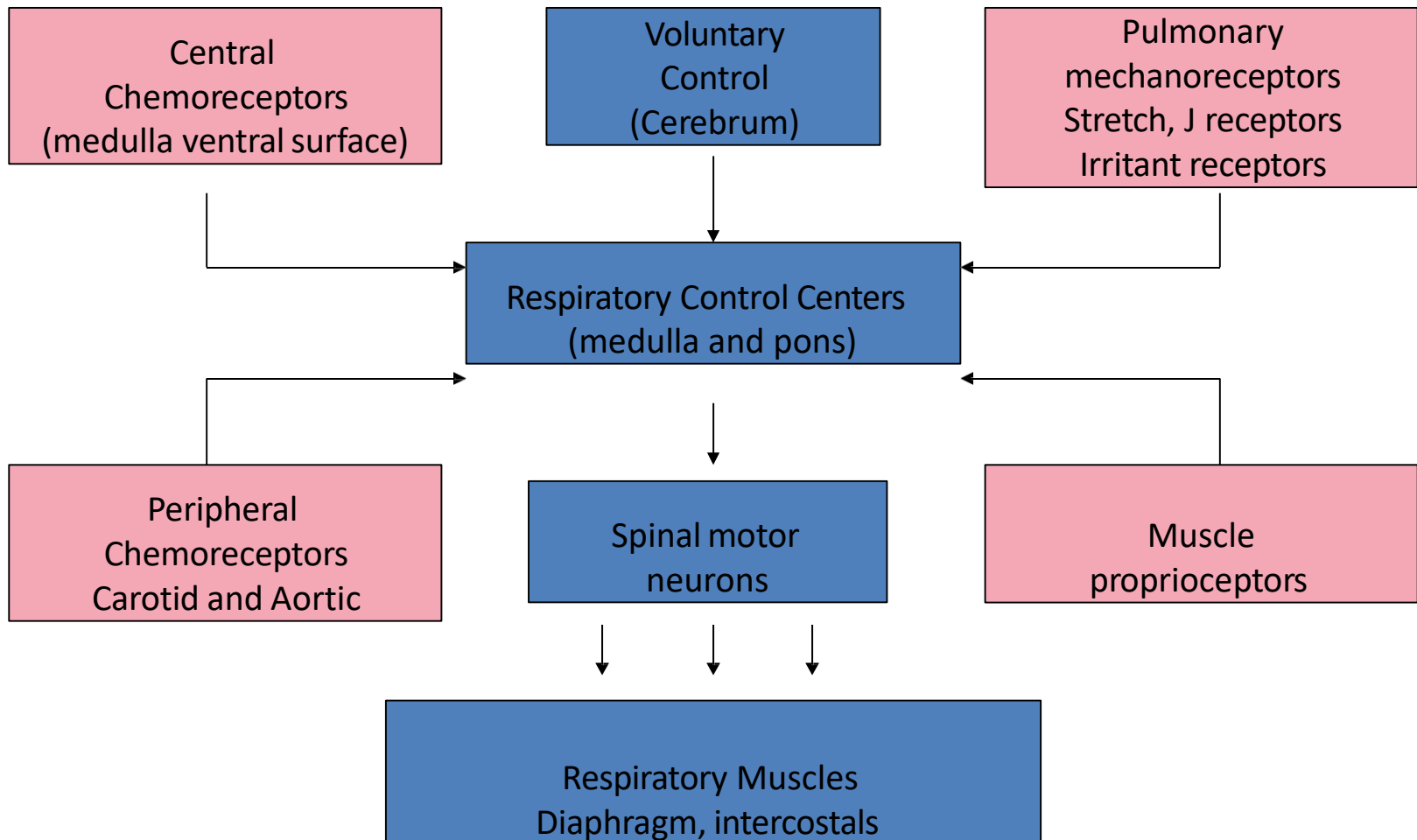


Fig. 5.32 Brain stem control of breathing. Afferent (sensory) information reaches the medullary inspiratory center via central and peripheral chemoreceptors and mechanoreceptors. Efferent (motor) information is sent from the inspiratory center to the phrenic nerve, which innervates the diaphragm. *CN*, Cranial nerve; *DRG*, dorsal respiratory group.

Basic Elements

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CNS – Important Centers- recap

- Cerebrum – voluntary control
- Medulla oblongata – site of dorsal respiratory center and ventral respiratory center – generate the basic rhythmic pattern of breathing
- Pons – apneustic and pneumotaxic centers can modulate the basic pattern of the medulla but are not essential (historical)
- Emotional responses – limbic system and hypothalamus (anxiety, rage, fear)

Pulmonary Reflexes

Reflex Pathways

- **Mechanoreceptors**

- Pulmonary Stretch Receptors
 - Lie in the smooth muscle layer of the airways
 - Fire in proportionate response to the transmural pressure
- Irritant Receptors
 - Located in the airway epithelium
 - Respond to touch or noxious substances e.g. smoke / particle deposition and also to inflammatory histamines, serotonin and prostaglandins – i.e. activated during inflammation
 - Lung edema also stimulates these receptors
 - Stimulation results in coughing / gasping
- J Receptors
 - Juxtapulmonary capillary receptors (also called C-fiber endings) in Alveolar and bronchial groups
 - Fire in response to lung injury, overinflation, pulmonary edema, pulmonary embolism – but are **not sensitive** to inflammatory mediators
 - Bronchial C-fibers – **are sensitive** to inflammatory mediators
 - Results in rapid shallow breathing, bronchoconstriction, airway secretion

- **Muscle Proprioceptors**

- Present in joints, tendons and muscle – inform the brain of the position of the body

Reflex Pathways/Receptors

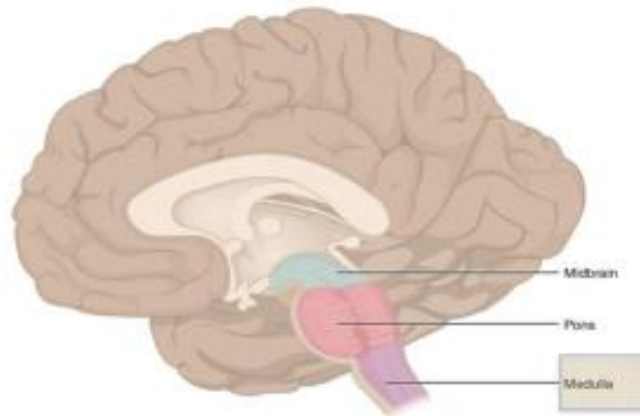
Reflex	Stimulus	Receptor	Afferent path	Effect
Cough	mechanical/ chemical irritation	Upper airway irritant receptors	Vagus	Cough, bronchoconstriction
Sneeze	mechanical/ chemical irritation	Nasal irritant receptors	Trigeminal, olfactory	Sneeze
Hering-Breuer inflation reflex	Lung inflation	Airway smooth muscle Stretch receptor	Vagus	Stops inspiration
Hering-Breuer deflation reflex	Lung deflation	Possible J, irritant or stretch receptors	Vagus	Starts inspiration
Positional Perception	Stretch of muscle or tendons	Proprioceptors	many spinal pathways	allows fine control of muscle groups
Pulmonary Vascular Congestion	edema	J receptors	Vagus	Tachypnea, sensation of dyspnea

Central Chemoreceptors and control of breathing

Sensors 1-- Central chemoreceptors

- Located in the ventral surface of the Medulla
- Responds to a change in PCO_2 and pH of the Cerebrospinal Fluid (CSF)

CSF is separated from the blood by the blood-brain barrier – which is **largely impermeable to H^+ and HCO_3^-**



Therefore, **metabolic acidosis or alkalosis has relatively little** effect of CSF pH.

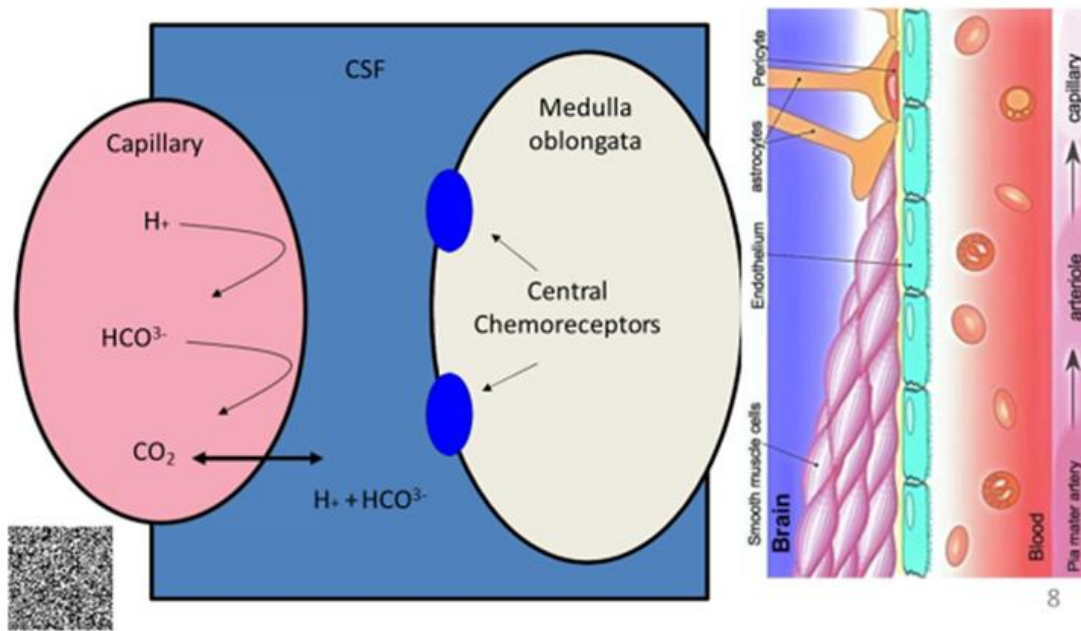
The blood-brain barrier is **very permeable to O_2 and CO_2 though** – therefore **PCO_2** has a strong effect on CSF pH

Cerebrospinal Fluid

CSF is very sensitive to changes in PCO_2 . Raising PCO_2 (at least initially) gives larger decreases in CSF pH.

Blood-Brain Barrier

CSF is separated from the blood by the blood-brain barrier – which is **largely impermeable to H^+ and HCO_3^-**



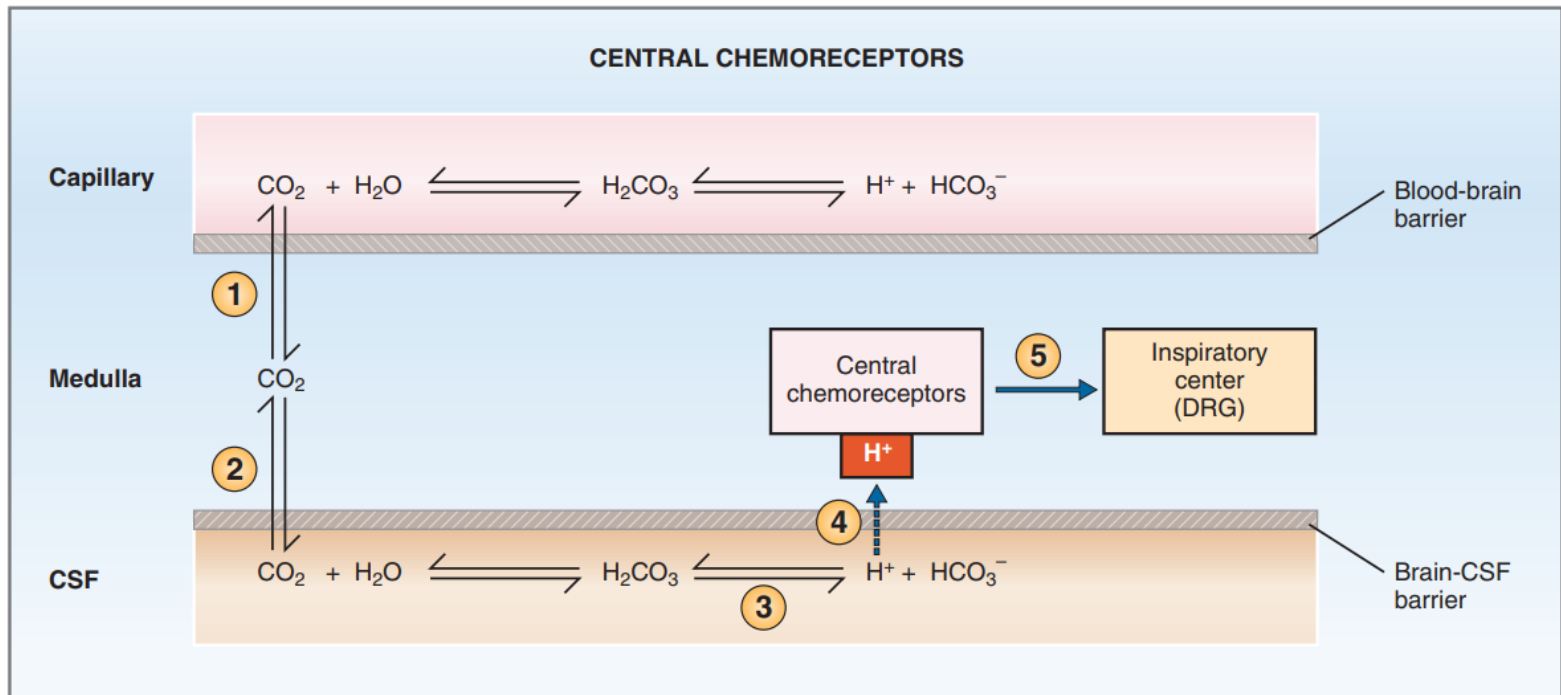
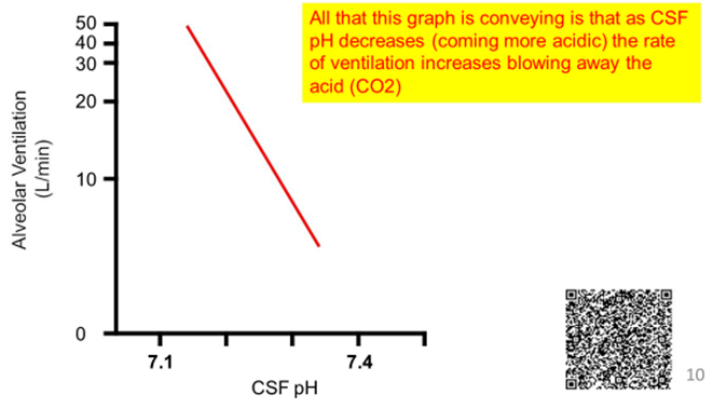


Fig. 5.33 Response of central chemoreceptors to pH. The circled numbers correspond to the numbered steps discussed in the text. *CSF*, Cerebrospinal fluid; *DRG*, dorsal respiratory group.

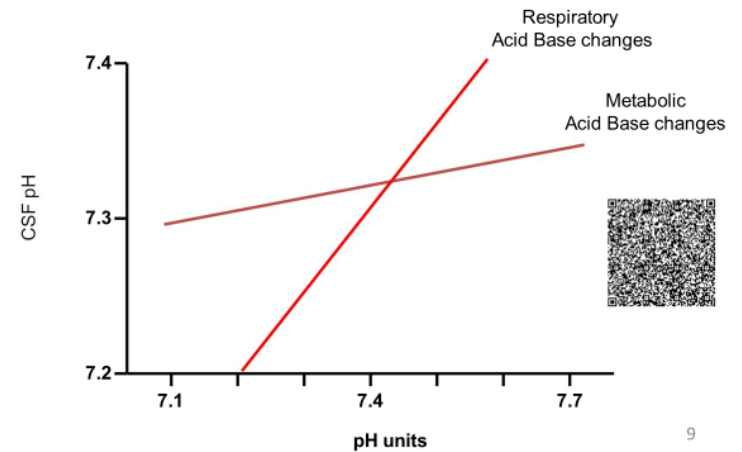
Costanzo Physiology ,7th Edition , Chapter 5 (Clinical Key)

PCO₂ changes pH and pH determines ventilation



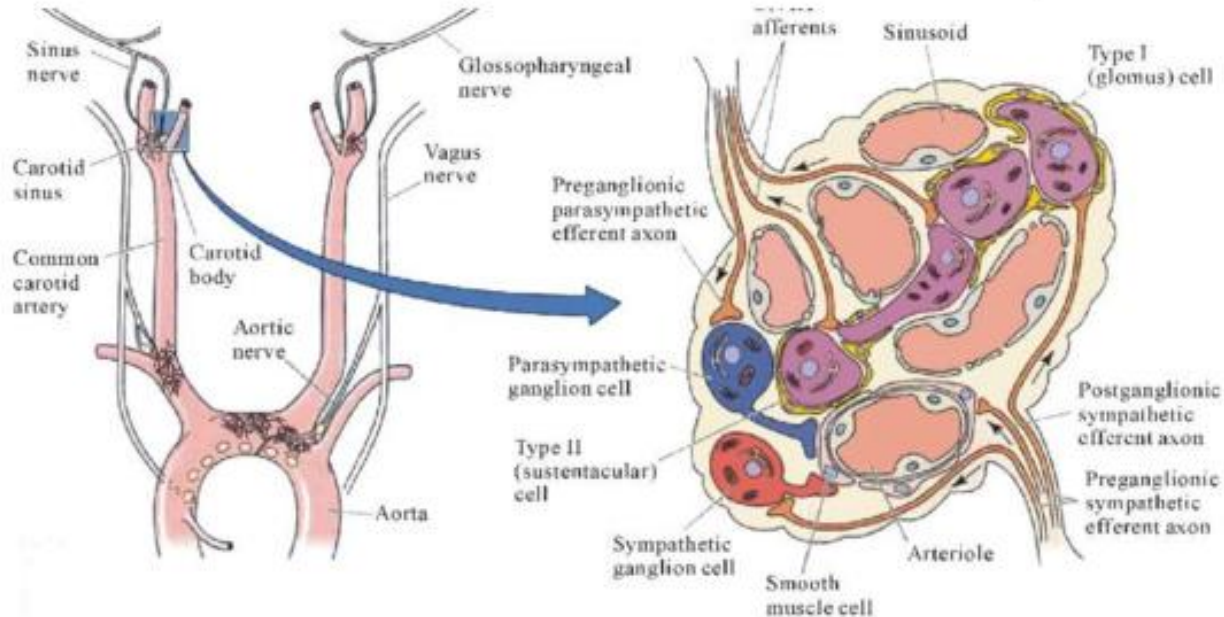
CSF pH

All that this graph is conveying is that CSF pH change can be affected only with respiratory acid base changes.



Peripheral Chemoreceptors

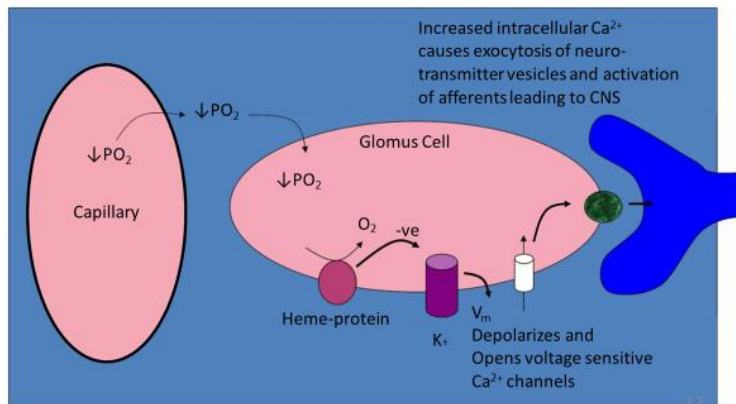
Sensors 2-- Peripheral Chemoreceptors



- Located in the carotid and aortic bodies
- Detect changes in arterial PCO_2 , PO_2 and pH
- Carotid body – small (2mm) sensory organ at the bifurcation of the carotid artery – afferents feed CNS via the glosso-pharyngeal nerves
- **Glomus cells – site of chemoreception**
- Aortic bodies – multiple bodies along aorta – afferents feed CNS via vagus nerve

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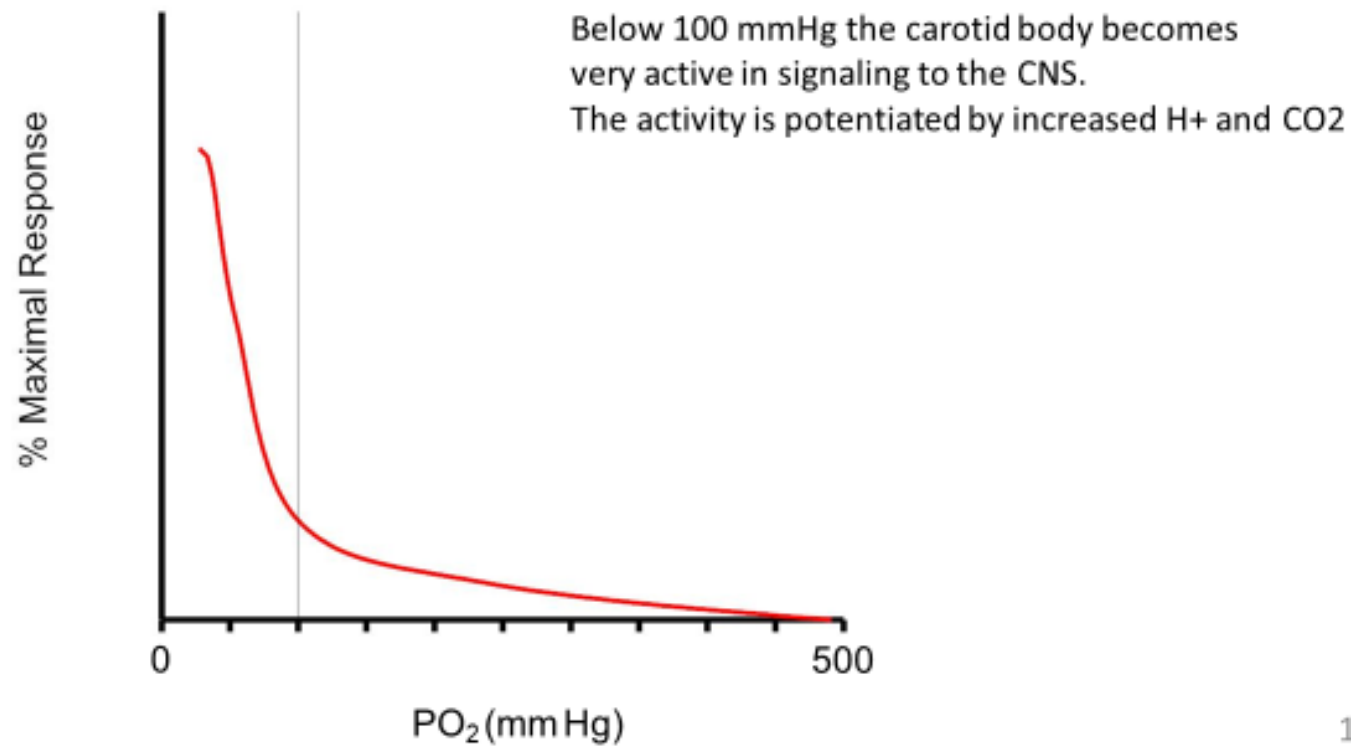
The Glomus Cell Hypoxic Response



The Glomus Cell Hypoxic & other Responses

- $\downarrow PO_2$ – depolarizes glomus cell and stimulates afferents to the CNS
- $\uparrow PCO_2$ in the cells causes acidification – which also causes depolarization
- H^+ - causes acid loading into cell - depolarization

Carotid Body



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Clinical Correlations

Clinical Case

A 40-year-old man is brought to the respiratory clinic because of a sudden and severe difficulty to breathe and chest tightness. He had a 30-pack-year smoking history and denied any illicit drug use. History revealed that he had worsening dyspnea on mild exercise and now even has dyspnea at rest. During the previous 12 months, he has had three hospital admissions for COPD exacerbations. Physical examination shows a dyspneic patient, using his accessory muscles while breathing and a significant ankle swelling. Auscultation shows decreased breath sounds and bilateral expiratory wheezing. X-ray shows hyperinflated lung fields with attenuation of peripheral vasculature, flattened diaphragms and increased retrosternal clear space, findings indicative of emphysema. ECG shows sinus tachycardia with no other remarkable findings. Spirometry demonstrated airflow obstruction with an FEV1. The arterial blood gas results are: pH 7.49; arterial carbon dioxide tension (P_a,CO_2) 3.9 kPa (29 mmHg); arterial oxygen tension (P_a,O_2) 6.4 kPa (48 mmHg); bicarbonate 26 mM; and arterial oxygen saturation (S_a,O_2) 84% on room air.

Research Questions

- 1. Understand the primary respiratory drive in this patient.
- 2. How will you treat this patient's breathlessness?
- 3. What is a possible regimen? 100% Oxygen administration or titrated oxygen?
- 4. What is the reasoning of this treatment regimen?
- 5. What are the potential consequences of administration of 100% Oxygen, in this patient?
- 6. In considering the treatment of this patient what are the factors to be taken into account?

Answers:

1. Hypoxic drive. Hypoxemia sensed by the **peripheral Chemoreceptors** is the only stimulus for the respiratory center to make this patient breathe. Central chemoreceptors are less responsive due to buffering changes in CSF by prolonged exposure to CO₂.
2. We must correct hypoxemia by giving supplemental oxygen or else the patient might have end organ damage due to hypoxia.
3. He must be treated with titrated oxygen and NOT 100% oxygen inhalation. The goal is to maintain the PaO₂ at 60-65 mm Hg and to achieve saturations of 88% to 92%.
4. This treatment regimen is to avoid consequences of harmful effects of 100% oxygen inhalation.
5. Potential consequences of 100% oxygen administration is retention of CO₂ in the body and its harmful effects in the organs. There are 3 important factors that contribute to this CO₂ retention as follows:
 - a) Due to delivery of abundant oxygen, the ventilation driven by hypoxemia through the peripheral chemoreceptors gets blunted. As this is the only drive for ventilation in him, the elimination of CO₂ suffers.
 - b) CO₂ transport from tissues to the lungs depends on deoxygenation of hemoglobin. This becomes defective with excess administration of oxygen.
 - c) The hypoxic vasoconstriction usually compensates the V/Q mismatch in these patients. However, administering excess oxygen removes hypoxic vasoconstriction which leads to higher V/Q mismatch and higher arterial CO₂.

Conclusion: In patients with an acute exacerbation of COPD, **do not administer 100% oxygen** to treat their hypoxemia. This may risk development of oxygen-induced hypercapnia. Therefore, titrate oxygen delivery to maintain the PaO₂ at around 60-65 mm Hg and to achieve saturation of 88% to 92%.

Clinical Case Reference Links

[Oxygen therapy and inpatient mortality in COPD exacerbation \(bmj.com\)](#)

[Use of Oxygen Therapy in COPD. Advantages and Information | Patient](#)

Any Questions ?

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